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$$\begin{aligned}\Delta E_{3-1} &= -1.51 - (-13.6) \\ &= 12.09 \text{ eV}\end{aligned}$$

$$\frac{hc}{\lambda} = \Delta E_{3-1} = 12.09 \text{ eV}$$

$$\Delta E_{\infty-1} = 0 - (-13.6) = 13.6 \text{ eV}$$

$$\begin{aligned}\frac{\Delta E_{\infty-1}}{\Delta E_{3-1}} &= \frac{13.6}{12.09} = \frac{\frac{hc}{\lambda_{\text{new}}}}{\frac{hc}{\lambda}} = \frac{\lambda}{\lambda_{\text{new}}} \\ \lambda_{\text{new}} &= \frac{12.09}{13.6} \lambda \approx \frac{8}{9} \lambda\end{aligned}$$

